China’s latest rare earth export controls add new twist in US chip war, observer says

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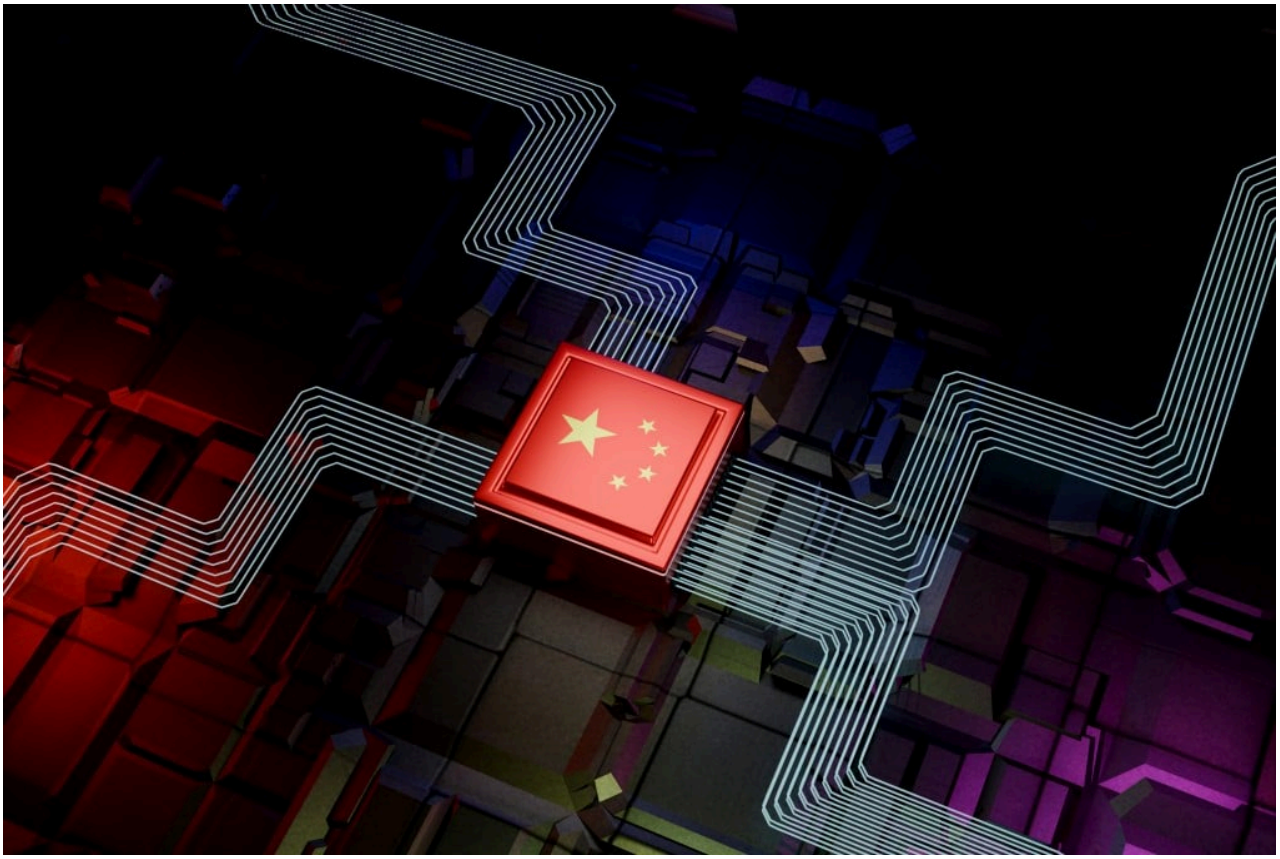
Ling Xin October 10, 2025



Ling Xin in Ohio
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China’s latest rare earth export controls could open an unexpected new front in its chip war with the United States, a prominent Chinese industry observer said.

Telecoms industry veteran Xiang Ligang said he was “very surprised” to see that new export control rules announced by the commerce ministry on Thursday included a clause requiring case-by-case review of rare earth exports intended for use in advanced chipmaking or military-related artificial intelligence (AI).

In a post on social media, Xiang said the clause suggested China had largely resolved its semiconductor equipment bottlenecks. He said it also indicated Beijing was unafraid of any major impact on foreign makers of advanced lithography systems that depend on Chinese rare earths.

“China’s chip industry no longer faces research and development hurdles – those have already been solved,” he wrote, adding that the next step was to expand production capacity and strengthen manufacturing.

He said some Chinese companies had reached world-class levels with equipment developed in just a couple years, and that Beijing could expand export controls to chip-making machines in the future, targeting the US.

Observer Xiang Ligang says China’s chip industry “no longer faces research and development hurdles”. Photo: Baidu

“China will also need to regulate the export of chip equipment and cannot allow it to be exported freely. The United States wants to develop its own chip manufacturing, [but it] cannot casually use equipment from China,” he said. “A few years ago, if we had said this, some people might have laughed at us.”

Beijing’s new export control rules, taking effect on December 1, aim to tighten oversight of rare earth materials and technologies that could be used in sensitive industries.

“Export applications for end uses involving the research, development or production of logic chips at 14 nanometres and below, or memory chips with 256 layers and above – as well as related manufacturing equipment, testing equipment and materials – will be subject to case-by-case review,” the fourth clause of the rules states.

Applications involving the research and development of AI with potential military uses will also be reviewed individually.

Rare earths are a group of 17 metallic elements essential to electric vehicles (EVs), wind turbines and high-end electronics. China dominates the global supply, producing about 70 per cent of mined output and more than 80 per cent of refined materials.

In chipmaking, their role is indirect but critical. For instance, neodymium and dysprosium power the precision motors inside lithography machines, while lanthanum and cerium are used in the manufacture of optical glass and polishing compounds for their lenses.

Tightening controls on such materials could complicate efforts by foreign companies to produce or maintain the advanced tools used in cutting-edge semiconductor manufacturing.

The new rules add another twist to the years-long struggle for technological dominance between Beijing and Washington. The US began curbing China’s access to advanced chips and manufacturing tools in 2019, later extending those bans to AI processors and third-country suppliers that use US technology.
Beijing’s countermeasures on critical minerals now mark a shift from defence to offence in the global chip rivalry. The clause singles out two benchmarks that define today’s most advanced chip production: 14-nanometre logic chips and 256-layer 3D memory.

Chips at these levels power high-performance computing and AI systems – the same areas where the US has sought to limit China’s progress.

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What are rare earths, and why is China's dominance facing global pushback

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"We used to think China couldn't make semiconductor equipment and had to rely on foreign suppliers," Xiang wrote. "But we soon found that China has developed a full range of tools matching the world's best – and some will soon surpass them."

He said such statements might have drawn ridicule a few years ago. "Now these machines are being released and already running on production lines. China's chip manufacturing is quickly moving from mature to advanced processes," Xiang wrote.

He said that since 2021 – in less than five years – the country's semiconductor equipment bottlenecks had been basically resolved, adding that "even as an optimist, I didn't expect it to happen this fast".

The latest list of restricted items covers a range of rare earth metals and alloys, including neodymium, praseodymium, samarium and dysprosium, along with magnetic materials and sputtering targets made from these elements. Exports of related parts and components will also require government approval.

Also on Thursday, the commerce ministry and customs announced separate export controls on lithium battery and graphite anode materials, which are key components in EVs, as another step in China's broader effort to protect its strategic technologies.

Ling Xin

Ling Xin is a science journalist based in Ohio. She mainly covers physics, astronomy and space. Her writing has appeared in Science, Scientific American, MIT Technology Review and other English and Chinese outlets. She was a visiting journalist at Science magazine in Washington, and has a master's degree in journalism from Ohio University.